# Mean CI, σ known -> Z

C8.01-C8.68 CORE ONLY (SKILL IDs)

## 1 RECOGNISE IT

- Estimate population mean μ.
- Population σ is explicitly supplied/known.
- Random sample; population normal or sample large enough.

## 2 DECISION / METHOD

1. Compute $\bar{X}$ and n.
2. SE=σ/√n.
3. z*=NORM.S.INV(1-α/2) > 0.
4. $\bar{X}$±z*SE.
5. Report units and confidence.

## 3 FORMULA

$z^* = \Phi^{-1}(1-\alpha/2) > 0$

$CI: \bar{X} \pm z^* \times \sigma/\sqrt{n}$

## 4 FORMULA SHEET VS WHAT I KNOW

Gives: $\bar{X}$, SE=σ/√n, Z lane when σ known. Still know: critical value, CI endpoints and interpretation.

## 5 EXCEL ROUTE

- `n=COUNT(range); Xbar=AVERAGE(range)`
- `zstar=NORM.S.INV(1-alpha/2)`
- lower/upper = `Xbar +/- zstar*sigma/SQRT(n)`.

## 6 TINY WORKED EXAMPLE

$\bar{X}$=40,σ=6,n=36,C=.95. SE=1 and z=1.96, so CI is `(38.04,41.96)`.

## 7 TOP EXAM TRAPS

- Using sample s when σ is supplied.
- Using t in the known-σ lane.
- Treating the CI as a range for individual X values.

## 8 SAY ALOUD

Known population σ means a Z interval for μ centred at $\bar{X}$ with SE σ/√n.

## 9 EXIT CHECK

What changes in the interval if n quadruples but everything else stays fixed?

**ONE-GLANCE VISUAL ROUTE**

